Proof Verification Dossier

Bounds and Estimators for the Civilian-to-Combatant Death Ratio

Internal document for manual checking — not for submission.

Machine companions: `analysis/verify_proofs.py` (sympy + Monte Carlo, all checks pass) and `formal/casualty_proofs` (Lean 4 + mathlib, kernel-checked, no `sorry`s).

August 25, 2026

How to read this document. Each claim is stated exactly as in the manuscript, followed by a proof written in more detail than the paper's appendix, an explicit list of the assumptions used at each step, and a verification note describing the symbolic (sympy) or Monte-Carlo check performed. Most content is additionally machine-checked in Lean 4 with mathlib (`formal/casualty_proofs`): Claims 1–6 in full (identification algebra, monotonicity and derivative, η-slack, manpower bound, delta-method gradients, Gauss–Markov optimality), the infimum and monotonicity ingredients of Claim 7, and the Section 6 arithmetic over exact rationals. The Lean development compiles with no unproven statements; the deterministic set-containment and multilinear-expansion statements of Claim 8 and the topological limit step rest on the symbolic and Monte-Carlo checks.

Contents

1 Setup and standing conventions **2**

2 Claim 1: benchmark identification (paper eq. 2, Assumption 3.1) **2**

3 Claim 2: sharp identified set (paper Theorem 3.3) **2**

4 Claim 3: combatants outside AM (paper Remark 3) **3**

5 Claim 4: manpower bound (paper Corollary 3.4) **4**

6 Claim 5: delta-method standard error (paper §3) **4**

7 Claim 6: precision-weighted aggregation (paper Proposition 2.1) **5**

8 Claim 7: feasibility and contradiction (paper Theorem 4.3) **5**

9 Claim 8: deterministic source-deletion sensitivity (paper Proposition 5.1) **6**

10 Section 6 arithmetic (spot summary) **6**

1 Setup and standing conventions

Population of size N; deaths D in the window $[t_0, t_1]$. Classes: W (women of all ages + males under 18) with population share w, and AM (*all* males 18+) with population share a, so

$$w + a = 1 \quad \text{exactly.}$$

The exact partition matters: the identification formulas substitute $a = 1 - w$, so any convention in which AM excludes part of the male population (e.g. men over 60) silently breaks them.

Combatant stock M, $f = M/N$, all combatants in AM (slack for exceptions handled in Claim 3). Estimand $q = D_M/D$, the combatant share of the dead; ω = share of the *dead* in class W.

The hazard model. All identification results use one structural model: conditional on being a civilian, a member of class W dies with per-capita hazard λ and a member of civilian-AM with hazard $\mu\lambda$, $\mu \geq 1$. Combatant deaths are governed separately (their total is qD by definition). No assumption is made about λ itself — it cancels.

2 Claim 1: benchmark identification (paper eq. 2, Assumption 3.1)

Claim 1 (Point identification under exchangeable collateral)**.** *If all civilians face equal per-capita hazard ($\mu = 1$), then*

$$q = 1 - \frac{\omega(1 - f)}{w}.$$

Proof. Civilian population masses: class W has wN civilians (no combatants in W); civilian AM has $(a - f)N$. With equal hazard λ, expected civilian deaths in W and civilian-AM are λwN and $\lambda(a - f)N$, so the fraction of *civilian* deaths landing in W is

$$\frac{wN}{wN + (a - f)N} = \frac{w}{w + a - f} = \frac{w}{1 - f},$$

where the last step uses $w + a = 1$. Combatant deaths (share q of all deaths) land entirely in AM. Hence the share of *all* deaths in W is

$$\omega = (1 - q) \cdot \frac{w}{1 - f},$$

and solving for q gives the display. Note this is exact in expectation (a first-moment identity), not an approximation; finite-sample noise is handled by the delta method (Claim 5). □

Verification status. **VERIFIED**. Sympy: solving $\omega = (1 - q)w\lambda/(w\lambda + (a - f)\mu\lambda)$ for q and substituting $\mu = 1$, $a = 1 - w$ reproduces the display exactly (λ cancels symbolically). Lean: `claim1_benchmark_identification`. Degenerate cases: $\omega = w$, $f = 0 \Rightarrow q = 0$; $\omega = 0 \Rightarrow q = 1$. *Subtlety for manual checking:* the last step of the proof needs $w + a = 1$ exactly (see the convention note in Section 1); with $a < 1 - w$ the two displayed forms of the identity disagree.

3 Claim 2: sharp identified set (paper Theorem 3.3)

Claim 2 (Sharp identified set under μ-bounded exposure)**.** *Suppose $f \leq a$ and the hazard model holds for some $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$. Given (ω, w, a, f),*

$$q(\mu) = 1 - \omega \, \frac{w + \mu(a - f)}{w},$$

is monotone decreasing in μ, and the identified set for q is exactly $[\, q(\mu_{\text{hi}}),\, q(\mu_{\text{lo}})\,] \cap [0, 1]$.

Proof. Step 1 (the curve). With hazards $(\lambda, \mu\lambda)$, expected civilian deaths are proportional to $\lambda wN + \mu\lambda(a-f)N$, so the civilian share landing in W is $w/(w+\mu(a-f))$ and

$$\omega = (1-q)\,\frac{w}{w+\mu(a-f)}. \tag{1}$$

Solving (1) for q gives $q(\mu)$.

Step 2 (monotonicity). $\partial q/\partial\mu = -\omega(a-f)/w \le 0$, with equality iff $a = f$ (no civilian adult men) or $\omega = 0$.

Step 3 (validity: every point of the demographic interval with $q(\mu^*) \in [0,1]$ is attainable). Fix any $\mu^* \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$ such that $q(\mu^*) \in (0,1)$. With (D, N) fixed, the observed W-class deaths pin $\lambda = \omega D/(wN)$. The theorem's feasibility conditions imply $\lambda \le 1$ and $\mu^*\lambda \le 1$, so the civilian class death counts do not exceed their populations. The resulting demographic allocation satisfies (1) with $q = q(\mu^*)$. Boundary points are attained directly: $q(\mu^*) = 1$ forces $\omega = 0$ and hence $\lambda = 0$; $q(\mu^*) = 0$ is attained with no combatant deaths. Hence $q(\mu^*)$ is consistent with the demographic data. The separate stock restriction $D_M \le M$ is imposed in Claim 4; before that intersection, continuity and monotonicity give $[q(\mu_{\text{hi}}), q(\mu_{\text{lo}})] \cap [0,1]$.

Step 4 (sharpness: nothing outside is attainable). Conversely, any admissible configuration has some $\mu \in [\mu_{\text{lo}}, \mu_{\text{hi}}]$, and (1) then forces $q = q(\mu)$, which lies in the interval. The intersection with $[0,1]$ imposes the logical range of a share. □

Verification status. **VERIFIED**. Sympy confirms the inversion and the derivative; Lean: `claim2_inversion`, `claim2_monotone`. 2,000 random parameter draws confirm endpoint ordering. The forward model's three death-shares (combatant, civilian-AM, W) sum to 1 symbolically. *Caveat worth knowing when checking manually:* the theorem treats ω as an exact population quantity; the "identified set" is a set for the estimand given error-free inputs. Sampling error enters separately via Claim 5 and systematic error via the anchor sensitivity analysis. This is stated in the paper's Discussion (limitation 3).

4 Claim 3: combatants outside AM (paper Remark 3)

Claim 3 (η-slack)**.** *Suppose a fraction $\eta \in [0, \bar\eta]$, $\bar\eta = 0.03$, of combatant* deaths *is recorded in class W, while the combatant stock stays in* AM *(a death-classification slack, so $S := w/(w+\mu(a-f))$ is unchanged). The observable identity becomes $\omega = (1-q)S + \eta q$, so*

$$q_{\text{true}} = \frac{S-\omega}{S-\eta}, \qquad q_{\text{true}} - q_{\eta=0} \;=\; \frac{\eta(S-\omega)}{S(S-\eta)} \;=\; \frac{\eta\, q_{\text{true}}}{S} \;=\; \frac{\eta\, q_{\eta=0}}{S-\eta},$$

and the bias is ≥ 0 whenever $S > 0$ and $q_{\text{true}} \ge 0$: ignoring the slack understates *q. Provided $S > \bar\eta$ (which holds at the Gaza values, where $S \ge 0.626$ for $\mu \le 2$), a loose uniform bound over $\mu \in [1,2]$ with $q_{\eta=0} \le 0.33$ is $\bar\eta \cdot 0.33/(0.626-\bar\eta) \approx 1.7$ percentage points; the exact per-μ bias is smaller and decreasing in μ (1.30 points at $\mu = 1$, 1.10 at $\mu = 1.5$, 0.83 at $\mu = 2$), because $q_{\eta=0}$ falls with μ faster than S does.*

Proof. Deaths in W now come from two sources: civilian deaths landing in W, which is $(1-q)S$ of all deaths as in Claim 2, plus the fraction η of combatant deaths, contributing ηq. Setting $\omega = (1-q)S + \eta q = S - q(S-\eta)$ and solving for q gives $q_{\text{true}} = (S-\omega)/(S-\eta)$. Subtracting $q_{\eta=0} = (S-\omega)/S$,

$$q_{\text{true}} - q_{\eta=0} = (S-\omega)\Big(\frac{1}{S-\eta} - \frac{1}{S}\Big) = \frac{\eta(S-\omega)}{S(S-\eta)}.$$

Substituting $S - \omega = q_{\text{true}}(S-\eta)$ gives the second form $\eta q_{\text{true}}/S$; substituting $S - \omega = q_{\eta=0}S$ gives the third form $\eta q_{\eta=0}/(S-\eta)$.

Nonnegativity. From the second form, $q_{\text{true}} - q_{\eta=0} = \eta q_{\text{true}}/S \geq 0$ for $S > 0$, $\eta \geq 0$, $q_{\text{true}} \geq 0$. (This is the right route: arguing instead from "$\omega \leq S$" requires $S \geq \eta$, since $\omega = S - q(S - \eta)$ gives $\omega \geq S$ when $\eta > S$ and $q \geq 0$.)

Bound. For the uniform bound the third form is used with the worst admissible values: it is increasing in η (for $S > \eta$) and in $q_{\eta=0}$, and decreasing in S, so on $\eta \leq \bar{\eta}$, $q_{\eta=0} \leq 0.33$, $S \geq 0.626$ (the $\mu = 2$ value; S decreases in μ) it is at most $\bar{\eta} \cdot 0.33/(0.626 - \bar{\eta}) < 0.017$. This is loose because it pairs the smallest S (attained at $\mu = 2$) with the largest $q_{\eta=0}$ (attained at $\mu = 1$): the exact per-μ bias $\bar{\eta}\, q_{\eta=0}(\mu)/(S(\mu) - \bar{\eta})$ evaluates to 0.0130, 0.0110, 0.0083 at $\mu = 1, 1.5, 2$.

Note the distinction between the second and third forms: the bias equals $\eta q/(S - \eta)$ with q the *no-slack* estimate, and $\eta q/S$ with q the *true* share — the two coincide only to first order in η. □

Verification status. **VERIFIED**. Sympy: all three displayed forms are symbolically equal to $q_{\text{true}} - q_{\eta=0}$; the superficially plausible chain $\eta q_{\text{true}}/S \cdot S/(S - \eta)$ (which double-counts the $(S - \eta)^{-1}$ correction) is confirmed *not* to equal the bias. Lean: the identity `claim3_qtrue`, the three bias forms, nonnegativity exactly under $S > 0$, $\eta \geq 0$, $q_{\text{true}} \geq 0$ (theorem `claim3_bias_nonneg_of_qtrue`), the spurious-chain refutation, and exact rational bounds on the per-μ biases. Numeric: $S(1) = 0.770$, $S(1.5) = 0.691$, $S(2) = 0.627$ at Gaza demographics. Direction note: the slack makes the paper's *upper* bounds conservative in the claim-favourable direction, so no downstream conclusion depends on suppressing it.

5 Claim 4: manpower bound (paper Corollary 3.4)

Claim 4. *$q \leq M/D$; whenever $\max\{0, q(\mu_{\text{hi}})\} \leq \min\{1, q(\mu_{\text{lo}}), M/D\}$, the joint identified set is $[\max\{0, q(\mu_{\text{hi}})\},\ \min\{1, q(\mu_{\text{lo}}), M/D\}]$; otherwise it is empty and the inputs are jointly infeasible.*

Proof. Combatants killed cannot exceed combatants existing: $D_M \leq M$. Divide by $D > 0$. The joint set is the intersection of two intervals, hence the endpoint formulas. (If the intersection is empty, the inputs are jointly infeasible — this is exactly the event detected by a positive contradiction radius.) □

Verification status. **VERIFIED** (algebraic identity; Lean: `claim4_manpower_bound`). One caveat for manual review: M here is the stock at t_0; if recruitment during the war is material, M should be the stock-plus-inflow, which *raises* the bound. In the Gaza application the bound is non-binding ($M/D = 0.630$ vs. the demographic cap $q(1) = 0.321$).

6 Claim 5: delta-method standard error (paper §3)

Claim 5. *For $\hat{q} = 1 - \hat{\omega}(1 - \hat{f})/\hat{w}$ with independent inputs,*

$$\widehat{\text{Var}}(\hat{q}) \approx \frac{\hat{\omega}^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{f}) + \frac{(1-\hat{f})^2}{\hat{w}^2}\widehat{\text{Var}}(\hat{\omega}) + \frac{\hat{\omega}^2(1-\hat{f})^2}{\hat{w}^4}\widehat{\text{Var}}(\hat{w}).$$

Proof. First-order Taylor expansion of $g(\omega, f, w) = 1 - \omega(1 - f)/w$:

$$\partial_f g = \frac{\omega}{w}, \qquad \partial_\omega g = -\frac{1-f}{w}, \qquad \partial_w g = \frac{\omega(1-f)}{w^2}.$$

$\text{Var}(\hat{q}) \approx \sum(\partial g)^2 \text{Var}$, using independence (no covariance terms). Squaring the gradients gives the display. □

Verification status. **VERIFIED**. All three partial derivatives and the assembled formula verified symbolically; the gradients are also kernel-checked as `HasDerivAt` statements in Lean (`claim5_grad_f/om/w`). Assumption to flag for manual review: independence of $(\hat{\omega}, \hat{f}, \hat{w})$ is plausible here because they come from different institutions (deaths records, military-balance estimates, census), but any common-mode error (e.g. both ω and w derived from the same census frame) would add covariance terms. The paper's conservative $\sigma_\omega = 0.01$ (5× the binomial SE) partially absorbs this.

7 Claim 6: precision-weighted aggregation (paper Proposition 2.1)

Claim 6. *For independent unbiased sources with known variances σ_i^2 (with estimated $\hat{\sigma}_i^2$ independent of the point estimates the plug-in combination remains conditionally unbiased but not in general minimum-variance in finite samples; when the estimates are also consistent it is asymptotically efficient among linear unbiased combinations, under standard regularity), the estimator*

$$\hat{\theta}_{\text{agg}} = \frac{\sum_i \hat{\theta}_i/\hat{\sigma}_i^2}{\sum_i 1/\hat{\sigma}_i^2}$$

is minimum-variance among linear unbiased combinations with weights fixed given the variances. With per-source biases b_i, the aggregate bias is the precision-weighted mean of the b_i.

Proof. (i) Minimise $\sum c_i^2\hat{\sigma}_i^2$ subject to $\sum c_i = 1$. The Lagrangian condition gives $2c_i\hat{\sigma}_i^2 = \kappa$, i.e. $c_i \propto 1/\hat{\sigma}_i^2$; the constraint fixes the normalisation. The objective is strictly convex, so this is the unique minimum (Gauss–Markov in one dimension).

(ii) $\mathbb{E}\hat{\theta}_{\text{agg}} - \theta = \sum_i c_i b_i$ with $c_i = \hat{\sigma}_{\text{agg}}^2/\hat{\sigma}_i^2$; direct substitution. □

Verification status. **VERIFIED**. Optimality confirmed on 300 random configurations against random competitor weights; bias formula confirmed symbolically. In Lean, the precision weights are proved to sum to one and the minimum-variance inequality $\sum c_i^2\sigma_i^2 \geq 1/\sum \sigma_i^{-2}$ is proved for every unbiased combination via Cauchy–Schwarz (`claim6_optimality`).

8 Claim 7: feasibility and contradiction (paper Theorem 4.3)

Claim 7. *Fix $(\hat{D}, \hat{N})$ and $[\mu_{\text{lo}}, \mu_{\text{hi}}]$. For each independent input $A \in \{\omega, w, M\}$, revise that input alone, re-impose $a' = 1 - w'$ when w moves and $f' = M'/\hat{N}$ when M moves, and define*

$$\rho_A^{(i)} := \inf\left\{\frac{|A' - \hat{A}|}{\sigma_A} : q^{(i)} \in \mathcal{F}(\hat{x}[A \mapsto A'])\right\}, \qquad \rho^{(i)} := \min_A \rho_A^{(i)}.$$

Then (i) $\rho^{(i)} = 0$ iff $q^{(i)} \in \mathcal{F}(\hat{x})$, and (ii) every one-axis rescue on axis A moves that input by at least $\rho_A^{(i)} \geq \rho^{(i)}$ standardised units.

Proof. (i, ⇐) If $q^{(i)} \in \mathcal{F}(\hat{x})$, the unmoved value $A' = \hat{A}$ is feasible on every axis, so every axis radius and hence $\rho^{(i)}$ is zero.

(i, ⇒) If $\rho^{(i)} = 0$, some axis A has $\rho_A^{(i)} = 0$. Take feasible revisions $A'_n \to \hat{A}$. The re-imposed accounting identities converge with that axis. Each revision carries some μ_n in the compact exposure interval; pass to a subsequence with $\mu_n \to \mu^*$. Continuity of the demographic equation (with $w > 0$) and of the manpower inequality gives both membership conditions of $\mathcal{F}(\hat{x})$ in the limit. Thus $q^{(i)} \in \mathcal{F}(\hat{x})$.

(ii) Any feasible one-axis revision is an element of the set over which $\rho_A^{(i)}$ takes its infimum, so its standardised displacement is at least $\rho_A^{(i)}$, which is in turn at least $\min_B \rho_B^{(i)} = \rho^{(i)}$. □

Verification status. **VERIFIED**. Kernel-checked ingredients (Lean): feasibility is monotone in M; ω_{needed} is antitone in μ; the radius lower-bounds every rationalisation's penalty and vanishes at a feasible face-value point. The compactness/continuity limit step is proved in the manuscript appendix. The code computes the ω-axis radius as the distance from $\hat{\omega}$ to $[\omega_{\text{needed}}(\mu_{\text{hi}}), \omega_{\text{needed}}(\mu_{\text{lo}})]$, thereby granting the claim its most favourable admissible μ.

9 Claim 8: deterministic source-deletion sensitivity (paper Proposition 5.1)

Claim 8. *Let $\Theta_\varnothing = [L_0, U_0]$ be the compact face-value identified set. Deleting sources in T gives a compact set $\Theta_T \supseteq \Theta_\varnothing$ of diameter R_T; every nonempty identified set $\widetilde{\Theta}_T$ obtained by replacing those sources while retaining all others satisfies $\widetilde{\Theta}_T \subseteq \Theta_T$. Then (i) $\widetilde{\Theta}_T \subseteq [L_0 - R_T, U_0 + R_T]$. (ii) If d_T^- and d_T^+ are the worst downward and upward endpoint displacements over replacements, their deterministic multilinear extensions*

$$\Delta_s(\varepsilon) = \sum_{T \subseteq \mathcal{I}} \prod_{i \in T} \varepsilon_i \prod_{j \notin T} (1 - \varepsilon_j) d_T^s \quad (s \in \{-, +\})$$

satisfy $\Delta_s(\varepsilon) = \sum_i \varepsilon_i d_{\{i\}}^s + O(\|\varepsilon\|_\infty^2) \le \sum_i \varepsilon_i R_i + C\|\varepsilon\|_\infty^2$ for a finite constant C determined by the deletion sets. Hence $[L_0 - \sum_i \varepsilon_i R_i, U_0 + \sum_i \varepsilon_i R_i]$ is a conservative deterministic first-order sensitivity band, while part (i) governs a specified finite deletion pattern.

Proof. (i) Since $L_0, U_0 \in \Theta_T$ and every $x \in \widetilde{\Theta}_T$ also lies in Θ_T, both $|x - L_0|$ and $|x - U_0|$ are at most $\operatorname{diam}(\Theta_T) = R_T$. This gives the stated containment and, after taking endpoint and replacement suprema, $0 \le d_T^s \le R_T$.

(ii) Separate the singleton terms in the defining polynomial. For $T = \{i\}$, the coefficient is $\varepsilon_i \prod_{j \ne i}(1 - \varepsilon_j) = \varepsilon_i + O(\|\varepsilon\|_\infty^2)$. Every term with $|T| \ge 2$ is $O(\|\varepsilon\|_\infty^2)$; there are finitely many and $d_T^s \le R_T < \infty$. With $n = |\mathcal{I}|$, $\delta = \|\varepsilon\|_\infty$, and $R_{\max} = \max_{T \ne \varnothing} R_T$, the remainder is bounded in absolute value by $[n(n-1) + 2^n] R_{\max} \delta^2$. Thus $\Delta_s = \sum_i \varepsilon_i d_{\{i\}}^s + O(\delta^2)$, and $d_{\{i\}}^s \le R_i$ gives the conservative linear bound. Joint-removal diameters need not be bounded by sums of singleton diameters, so the finite-pattern claim remains part (i), not the first-order band. □

Verification status. **VERIFIED**. The containment is exercised on 1,000 randomly generated nested intervals and replacement subsets. The directional derivative of the multilinear deletion functional at the origin is checked in 100 random configurations, together with its singleton-diameter bound. Downstream use in Section 6 labels the quoted 32.1% → 39.4% inflation as deterministic first-order source-deletion sensitivity.

10 Section 6 arithmetic (spot summary)

All application numbers are generated by `analysis/validate_bounds.py`. Key identities are re-verified here at $(w, a, f, \omega) = (0.755, 0.245, 45000/2226544, 0.523)$; the starred rows are also proved over exact rationals in Lean.

Quantity	Value	Check
$q(1)$	32.13%	symbolic + numeric + Lean* (exact: 67511651/210130090)
$q(1.5)$	24.34%	numeric + Lean*
$q(2)$	16.56%	numeric + Lean*
root of $q(\mu) = 0$	$\mu^* = 3.06$	numeric; matches the q=0 symmetry discussion
μ needed for $q = 17{,}000/71{,}444$	1.54	inversion; ≥ 1 so arithmetically feasible
μ needed for $q = 25{,}000/71{,}444$	0.82	inversion; < 1 so infeasible for any admissible μ
M/D	0.630	non-binding vs. all bounds above
η-slack bias, μ=1, 1.5, 2	1.30, 1.10, 0.83 pp	numeric + Lean*; uniform bound 1.66 pp